

Masked Modeling-based Audio Representation for ACM Multimedia 2022 Computational Paralinguistics Challenge

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ABSTRACT

In this paper, we present our solution for ACM Multimedia 2022 Computational Paralinguistics Challenge. Our method employs the self-supervised learning paradigm, as it achieves promising results in computer vision and audio signal processing. Specifically, we firstly explore modifying the Swin Transformer architecture to learn general representation for the audio signals, accompanied with random masking on the log-mel spectrogram. The main goal of the pretext task is to predict the masked parts, by combining the advantages of the Swin-Transformer and masked modeling. For the downstream tasks, we utilize the labelled datasets to fine-tune the pre-trained model. Compared with the competitive baselines, our approach can provide significant performance improvements without ensembling.

CCS CONCEPTS

• Computing methodologies → Knowledge representation and reasoning; Machine learning.

KEYWORDS

Self-supervised Learning, Masked Modeling, Swin Transformer

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1 INTRODUCTION

The performance of audio signal classification heavily relied on the representation learning, in which the self-supervised learning paradigm can be helpful to generate representative and discriminative features [11], without requiring large-scale labelled datasets. A rich line of works has focused on self-supervised representation learning. In the early stage, researchers mainly focus on the designing of different pretext tasks, such as predicting a relative offset or the rotation angle, solving the jigsaw puzzles, inpainting and reconstruction. Due to its spectacular performance, self-supervised learning has become the main choice for different kinds of data in last two years.

In recent years, masked modeling [2, 4] has proven to be an effective method to improve the pre-training performance in natural language processing (NLP). These approaches mask a portion of the input sequence, such as a few words in a sentence and then train the models to predict these corrupted parts. Large-scale language models [15] are trained by numerous unlabeled text data have shown their good generalization to the number of downstream applications. In the computer vision field, contrastive predictive coding (CPC) [10, 18] predicts the patches using a verification task and contrastive predictive coding [14]. Vision Transformer (ViT) [5] masked patch prediction for self-supervised learning and Bidirectional Encoder representation from Image Transformers (BEiT) [1] proposes to predict the discrete tokens. Instead of considering the 2D image structure, Generative Pretraining from Pixels (iGPT) [3] directly predicts unknown pixels using a sequence Transformer. More recently, Masked Auto-Encoder (MAE) [8] predicts random masked patches of the input image and reconstructs the missing pixels by an autoencoder, and this method has obtained superior performance for the image classification task.

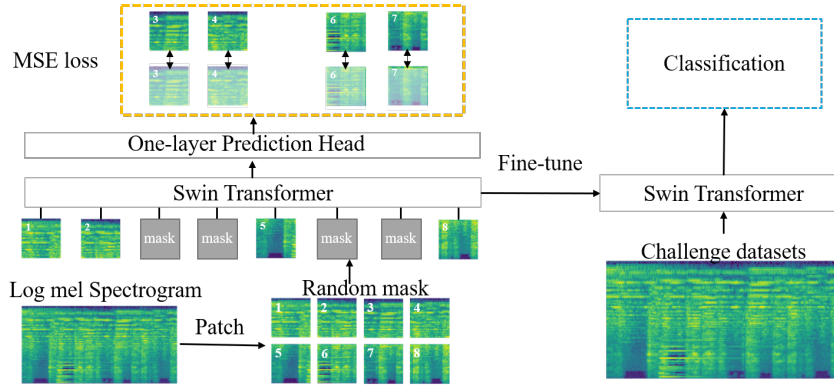


Figure 1: The framework of the proposed approach.

Despite the success of self-supervised learning in computer vision and NLP, the audio or speech signal is fundamentally different from natural language or natural images. Compared with the words in a sentence, the frames near each other are more correlative in the audio or speech signals, thus the pretext task may be well accomplished by copying the nearing frame but not by predicting them. Also, the text tokens are high-level concepts while the frames in audio are low-level signals. Even though the audio spectrogram can be regarded as a 2D image, it is still very different from the natural image. On the other hand, the event length can be diverse, and some short events are even only lasts a few milliseconds, which is challenging for representation learning. Audio spectrogram transformer (AST) [6] firstly uses a pure attention-based neural network for audio classification. This Transformer architecture needs more training data to compete with previous convolutional networks, and the pre-training phase has a great impact on the performance. After that, self-Supervised Audio Spectrogram Transformer (SSAST) [7] employs the masked modeling for the audio representation learning. It proposes a self-supervised method by reconstructing and classifying masked audio spectrograms. The network architectures of both aforementioned methods are similar to the ViT. It is not clear how to adapt the masked signal modeling approach to deal with audio and speech signals, and it is also unknown whether the learned representation can be helpful for the downstream applications, such as the data presented in the ACM Multimedia 2022 Computational Paralinguistics Challenge [16].

To answer aforementioned issues, in this paper, we present our exploration for the challenge leveraging the self-supervised learning paradigm. Specifically, we proposed an attention-based neural network to learn audio or speech representation with masked signal modeling. The backbone is built by an asymmetric encoder-decoder architecture. The encoder is a purely attention-based network to learn local and global self-attention in the audio spectrogram. The decoder could be flexibly designed and independent of the encoder. To reduce the number of weights and training time, our decoder is a lightweight network, while it can still retain a good performance. Additionally, we conduct the experiments with different mask strategies on the input log-mel spectrogram. All the learning processes do not require any human annotation of audio or speech. Finally, we evaluate our trained model on the challenge dataset by

fine-tuning the pre-trained model. Without ensembling, our solution can provide significant performance improvements, compared with competitive methods.

The remainder of this paper is organized as follows: Section 2 presents the methodology details while the experimental results on the challenge datasets are given in Section 3. Section 4 draws the conclusion.

2 METHODOLOGY

Figure 1 illustrates the overall pipeline of our solution. As shown in the figure, our methodology consists of two key modules: the pre-training phase on large-scale unlabeled datasets and the fine-tuning phase on different sub-challenge datasets. Our proposed method is motivated by the SimMIM [19], with the goal to combine the advantages of the Swin Transformer and masked modeling. In the following part of this section, we detail the structure of each component in our proposed framework.

2.1 Pre-training stage using Swin-Transformer and masked modeling

In our solution, we define the pretext task as a reconstruct task and utilizes a masking strategy to mask out a part of the log-mel spectrogram. Then, we use an encoder to extract the representation of input audio, and employ a decoder to reconstruct the masked part. This can encourage the model to learn the contextual information of the masked part, so that the model can extract representative and discriminative features.

Masking strategy: For the log-mel spectrogram of the audio signals, the mask strategy is mainly determined by three factors. The first factor is the masked shape. In our experimental settings, three different types of masked shapes can be used: time mask, frequency mask and patch mask, which are demonstrated in Figure 2. The second factor is the masked ratio, which represents the proportion of the part of the picture masked by the representative to all the log-mel spectrogram. The third factor is the mask shapes in the patch mask, which refers to the size of the mask patches. The patch mask focuses on encouraging the model to learn structural contextual features and perform better in the downstream tasks of audio tasks, while the time mask focuses on making the model

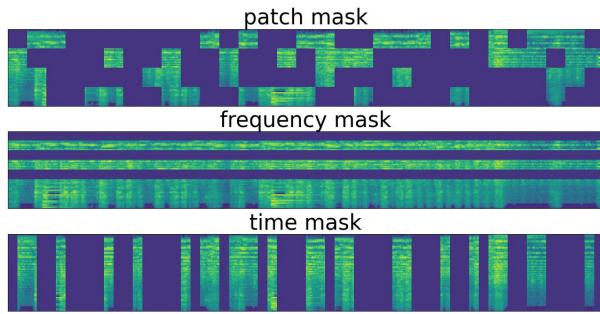


Figure 2: Masking strategy for the pre-training phase in our proposed method.

to learn time contextual features, which perform better in downstream tasks of speech tasks [7]. In our experiments, we employ the random patch mask strategy, while the mask ratio is set as 0.6 and the mask patch size is set as 32. In our experiments, we found that even with a very high mask ratio, our method can recover the masked parts with high precision, which can be demonstrated in the following parts.

Encoder: The encoder is used to extract the feature information of the log-mel spectrogram. In many previous attempts, people often use the convolutional neural network (CNN) as the backbone to learn the representation, such as VGG [17] and ResNet [9]. Recently, Transformers have become mainstream architectures in representation extraction and have achieved better performance in various downstream tasks with large-scale datasets, such as ViT [5], Swin-Transformer [12] in Computer Vision. As aforementioned, the researchers have proposed the AST [6] and SSAST [7] for audio signals. In our experiment, we use the modified Swin-Transformer as the backbone to extract audio representation, through the pre-training on the Audioset-2M dataset, which was shown in Figure 1.

Decoder: The input of the decoder consists of all visible tokens and mask tokens. Different from the MAE [8] which utilizes a multi-layer converter structure as decoder, we only employ a single-layer MLP is used as decoder. The lightweight decoder can greatly reduce the model training time without decreasing the accuracy. It is worth noting that: in our experiments, we observe that the model obtained using all tokens has poor performance with linear probing (such as SimMIM) settings, but this does not affect the final performance of fine-tuning. [7, 8].

Prediction head: First, the output vector of the decoder is mapped to the original resolution of the log-mel spectrogram. The vector performs the loss calculation with the original pixel value. We only calculate the loss on the mask tokens, which is consistent with MAE and SimMIM. We employ mean squared error (MSE) as the pixel reconstruction loss, and the reconstruction target is the normalized pixel value of each masked patch.

2.2 Fine-tuning for downstream task

After the pre-training stage, we firstly generate the log-mel spectrogram for the sub-challenge datasets. Then we use the pre-trained encoder to extract the representation of input audio, and adjust the

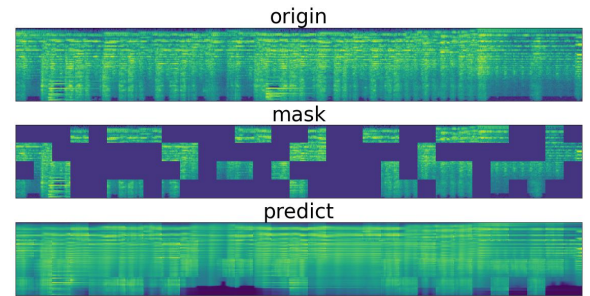


Figure 3: Reconstruction result using the pre-trained model.

parameters of the encoder. Finally, the classification head is used for audio classification tasks.

3 EXPERIMENTAL RESULTS

3.1 Implementation details

This part shows the implementation details of our solution for the challenge, which can be divided into two main parts. The first one is to provide the implementation details in pre-training phase. The second one aims to show the fine-tuning procedure on the two datasets VOC-C and KSF-C.

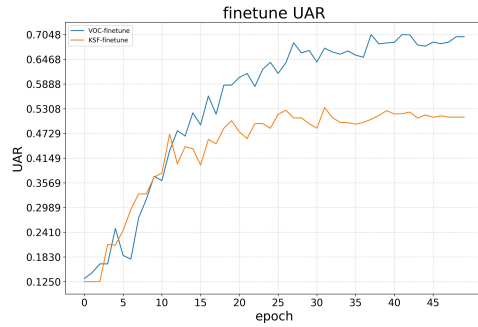
Pre-training: We use the Audioset training set for the pre-training task, which consists of 632 audio categories and 2,084,320 human-labeled 10-second sound clips from YouTube videos. Due to the expired links and network instability, a total of 1,911,936 audios were collected. The Audioset covers a wide range of human and animal sounds, instrumental and musical genre sounds, everyday ambient sounds. Therefore, using Audioset for pre-training can enable the model to learn various audio representations.

In terms of model structure, we use 4 Swin-Transformer blocks and the depths of each blocks are 2, 2, 18, 2 respectively. The number of Multi-Head of each block is set as 4, 8, 16, 32. The window size of the Swin-Transformer is set as 8. We firstly generate the log-mel spectrograms with the duration of 10 seconds, using the random mask method without any data augmentation. We input the log-mel spectrogram into the model to train 40 epochs. The warming-up epoch is set as 2, the learning rate is 0.003 while the batch size is 128. For the optimizer is, we select the AdamW [13], and the learning rate schedule is Cosine Decay Strategy in our experimental settings. We train the models for 72 hours with $8 \times$ NVIDIA A40 GPUs.

Fine-tuning: After the pre-training stage, we use the encoder parameters of the pre-trained model to initialize the classification model for the downstream task, so that the model can extract better audio representation for superior performance. For the data augmentation, the size of the frequency mask size and time mask is set as 48, and we add random noise to the data. We perform five-fold cross-validation on the train and eval dataset of VOC and KSF. In each fold, we train for 50 epochs, the warming-up epoch is set as 5, and the batch size is set as 32. The optimizer is adamw in our experimental settings, and the learning rate schedule is Cosine Decay Strategy. After the fine-tuning, we generate the probability of each category on the test dataset, and output the prediction result after averaging each probability.

Table 1: Results (%) on the VOC and KSF datasets.

Approach	VOC-C (UAR)		KSF-C (UAR)	
	Dev	Test	Dev	Test
DeepSpectrum	35.0	34.1	28.1	40.4
BoAWs	39.6	37.4	26.7	32.1
Scratch	25.0	\	29.0	\
SSAST	40.6	\	38.7	\
Ours	43.1	40.0	42.0	46.4

**Figure 4: Fine-tuning curve on the VOC-C and KSF-C datasets.**

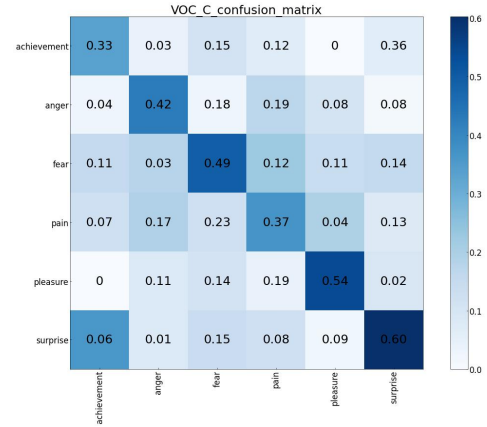
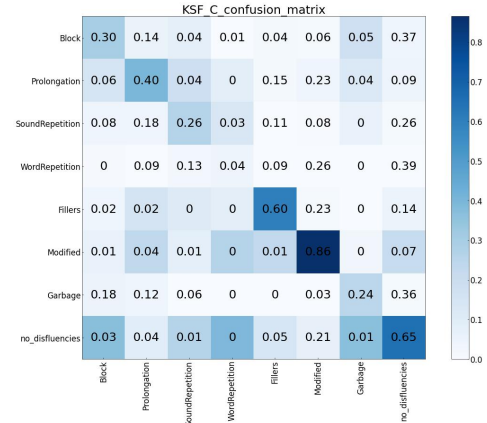
3.2 Visualization for the reconstruction

To demonstrate the performance of the pre-training, we firstly visualize the reconstructed results using the pre-trained model. As shown in Figure 3, the original log-mel spectrogram is displayed in the top row, while the random patch strategy is given in the second row. The reconstructed log-mel spectrogram is given in the bottom row. By visual observation, our pre-trained model can recover the corrupted log-mel spectrogram with high precision, even with a high mask ratio. These results may suggest that our proposed method can capture contextual information of the audio signals, which can be of great importance to representation learning.

3.3 Experimental results on the challenge datasets

For the challenge datasets, the training UAR curve in training is shown in the Figure 4. As shown in the figure, the UAR gradually converges after 40 epochs. The quantitative evaluation of the UAR is shown in the Table 1. As can be seen from the table, UAR is improved by 2.6% compared to the baseline on the VOC dataset, and 6% UAR improvement has been achieved on the KSF dataset compared to the baseline. The improvement of KSF is significant. Moreover, to further demonstrate the effectiveness of the pre-training process, we use random initialization parameters for the training from scratch, which can be used for the quantitative comparison.

As can be seen from the table, the performance of our proposed method is greatly improved, by 72.4% on the VOC dataset and 44.8% on the KSF dataset, compared with the training from scratch using

**Figure 5: Confusion matrix for VOC-C dataset.****Figure 6: Confusion matrix for KSF-C dataset.**

random initialization parameters. We also initialized the parameters using SSAST, which is the state-of-the-art audio representation. Compared with SSAST, our representation can also provide better performance on the develop datasets. The confusion matrices validated on Develop datasets are shown in Figure 5 and Figure 6. It is worth mentioning that: as our main goal is to improve the general representation ability of the audio signals, thus we don't fusion with other methods which may further boost the performance.

4 CONCLUSION

In this paper, we present our solution for the ACM Multimedia 2022 Computational Paralinguistics Challenge, leveraging the self-supervised learning paradigm. We firstly pre-trained a novel backbone, by combining the advantages of the Swin Transformer and masked modeling. Then we employ the labelled challenge datasets to fine-tune the pre-training model. Compared with state-of-the-art audio presentations (such as SSAST), our solution can provide better performance while requiring less computation cost. Without ensembling, compared with the baselines, our simple yet effective approach provides dramatic performance improvements on this challenge datasets.

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